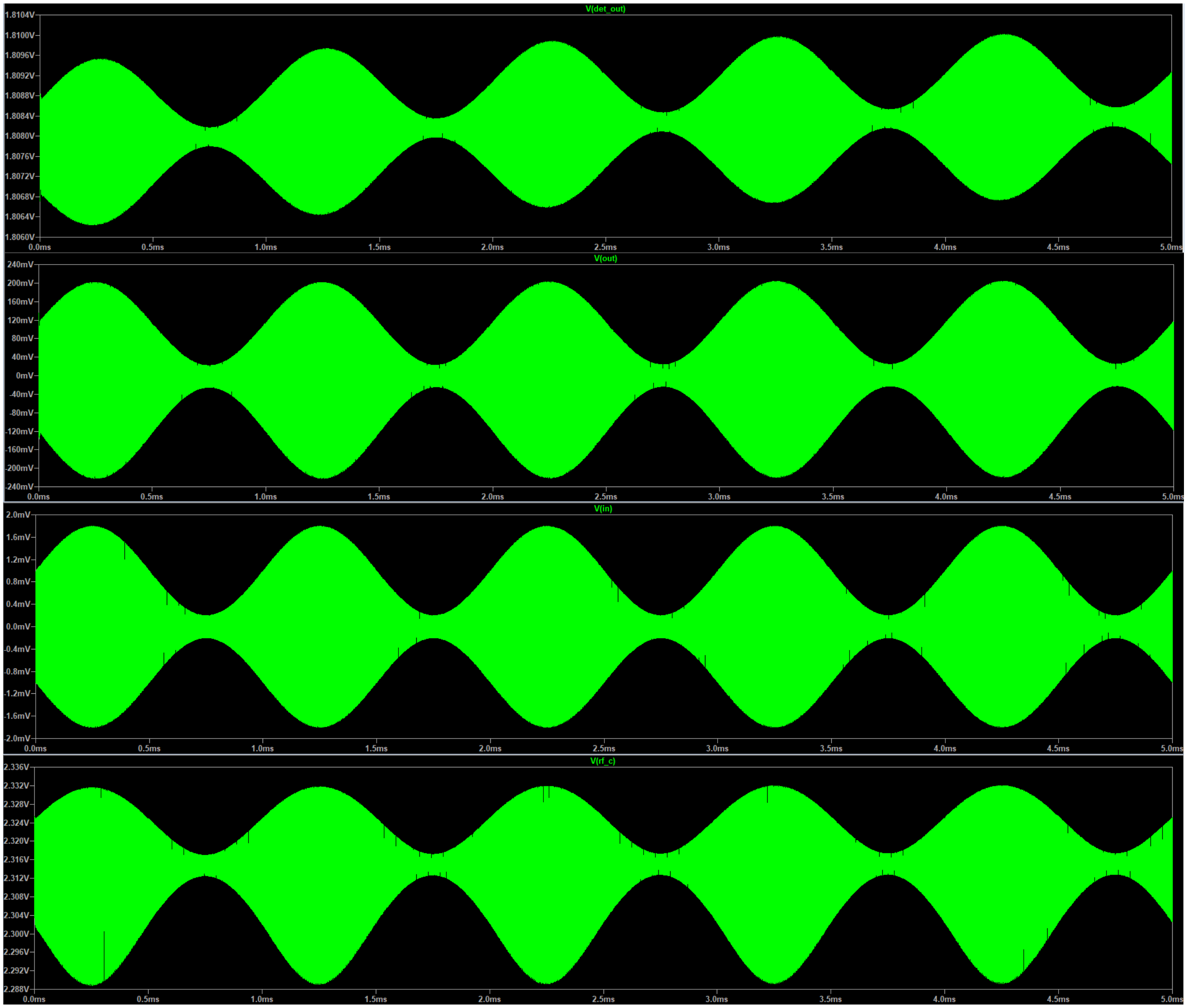
Real_LTSpice_Netlist_4

Prompt



here are the results i plotted

Answer

LTSpice AM Radio Receiver Project: Complete Log & Netlist

1. Project Requirements & Component List

Target Frequencies:

- 198 kHz (BBC Radio 4)

- 693 kHz (BBC 5 Live)
- 1467 kHz (Trans World Radio)

Physical Components:

- **Antenna:** Antenna wire (requires LC tuning circuit)
- **RF Stage:** 1× 2N3904 NPN transistor, 4.7 kΩ, 1 kΩ, 100 kΩ resistors, 2× capacitors (10nF – 100nF)
- **Detector Stage:** 1N4148 Diode, 10 kΩ resistor, 10 nF capacitor
- **Audio Stage:** 1× 2N3904 NPN transistor, 10 kΩ (base), 4.7 kΩ (collector), 1 kΩ (emitter) resistors, 10 μF coupling capacitors
- **Output / Power:** 9V battery, 32 Ω headphones

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2. LC Tuning Tank Calculations

To tune to specific stations, the LC tank (inductor L and capacitor C in parallel) acts as a bandpass filter.

The Resonant Frequency Formula:

$$f = \frac{1}{2\pi\sqrt{LC}}$$

$$C = \frac{1}{4\pi^2 f^2 L}$$

- **Tuning to 198 kHz (BBC Radio 4):**
  Using a 1 mH (0.001 H) inductor:

  $$C = \frac{1}{4\pi^2(198,000)^2(0.001)} \approx 6.46 \times 10^{-10}\,\mathrm{F}$$

  Result: 646 pF
- **Tuning to 693 kHz (BBC 5 Live):**
  Using a 220 μH (0.00022 H) inductor:
  Result: 240 pF
- **Tuning to 1467 kHz (Trans World Radio):**
  Using a 100 μH (0.0001 H) inductor:

  $$C = \frac{1}{4\pi^2(1,467,000)^2(0.0001)} \approx 1.17 \times 10^{-10}\,\mathrm{F}$$

  Result: 117 pF

3. Simulation & Verification Guide

To verify the circuit in LTSpice, run the simulation (`.tran 0 5m 0 100n`) and plot these critical nodes:

1. `V(in)` **(Antenna Input):** The raw AM signal. You should see a high-frequency carrier wave whose envelope grows and shrinks.
2. `V(rf_c)` **(RF Amplifier Output):** The exact same AM wave shape as `V(in)`, but amplified and riding on a positive DC voltage.
3. `V(det_out)` **(Detector Output):** The high-frequency carrier should be stripped away, leaving a smooth 1 kHz sine wave (the audio envelope) with a DC offset.
4. `V(out)` **(Final Audio Output):** A strong, clean 1 kHz sine wave perfectly centered at 0V, ready to drive the next audio stage.

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4. Troubleshooting Log

During simulation, three common analog design issues were encountered and resolved:

- **Issue 1: Flat Audio Output (Impedance Mismatch).** The audio transistor's 4.7 kΩ collector resistor creates a high output impedance. Connecting a tiny 32 Ω headphone load directly acts as a voltage divider, squashing the AC signal to zero.
  - *Fix:* Changed the simulation load `R_load` to 10 kΩ. (In reality, an LM386 chip is needed to drive 32 Ω headphones).
- **Issue 2: Squashed RF Wave (Clipping).** The initial 10mV antenna signal was too strong for the high-gain RF amplifier, causing the transistor to max out and clip the bottom of the wave flat.
  - *Fix:* Reduced input source `B1` to 1mV initially (later raised to 50mV after AC coupling the detector). Added an emitter bypass capacitor (`C_af_e` 10 μF) to the audio stage to regain volume.
- **Issue 3: Solid Green Blocks in Audio Stage (Permanently ON Diode).** The RF collector sits at a DC bias of roughly 2.3V. Because the 1N4148 diode was directly connected, the 2.3V kept the diode permanently turned "on" (it only needs 0.6V). It stopped rectifying and let the high-frequency RF flood into the headphones.
  - *Fix:* Added a 1 nF AC coupling capacitor (`C_couple`) between the RF stage and the diode to block the DC bias. Increased the antenna signal to 50mV to ensure the AC wave was strong enough to breach the diode's 0.6V threshold on its own.

5. Final Working LTSpice Netlist

Save the following code as `AM_Radio.cir` and run it in LTSpice to see the fully functioning 693 kHz receiver.

```
* AM Radio Receiver Simulation (693 kHz) - FINAL WORKING VERSION
* --- Power Supply ---
Vcc VCC 0 9V

* --- RF Antenna / Signal Source ---
* Amplitude set to 50mV to ensure enough AC swing to turn on the diode
B1 IN 0 V=(1+0.8*sin(2*pi*1k*time))*sin(2*pi*693k*time)*50m
R_ant IN RF_IN 50

* --- LC Tuning Tank (693 kHz) ---
L1 RF_IN 0 220u
C_tune RF_IN 0 240p

* --- RF Amplifier Stage (Common Emitter) ---
C_rf_in RF_IN RF_B 10n
R_rf_b RF_C RF_B 100k
R_rf_c VCC RF_C 4.7k
R_rf_e RF_E 0 1k
C_rf_e RF_E 0 100n     ; Emitter bypass for high RF gain
Q1 RF_C RF_B RF_E 2N3904

* --- AM Detector Stage ---
C_couple RF_C DET_IN 1n     ; Blocks DC offset from permanently activating
diode
D1 DET_IN DET_OUT 1N4148
R_det DET_OUT 0 10k
C_det DET_OUT 0 10n    ; 10nF balances the RC time constant perfectly

* --- Audio Amplifier Stage ---
C1 DET_OUT AF_B 10u
R_af_b AF_C AF_B 10k
R_af_c VCC AF_C 4.7k
R_af_e AF_E2 0 1k
C_af_e AF_E2 0 10u     ; Emitter bypass for high Audio gain
Q2 AF_C AF_B AF_E2 2N3904
C2 AF_C OUT 10u
R_load OUT 0 10k       ; 10k prevents simulation impedance mismatch

* --- Transistor & Diode Models ---
.model 2N3904 NPN(IS=1E-14 VAF=100 BF=300 IKF=0.4 XTB=1.5 BR=4 CJC=4p CJE=8p
RB=20 RC=0.1 RE=0.1 TR=250p TF=350p ITF=1 VTF=2 XTF=3 VCEO=40)
```

```
.model 1N4148 D(Is=2.52n Rs=.568 N=1.752 Cjo=4p M=.4 tt=20n Iave=200m Vpk=75)

* --- Simulation Setup ---
.tran 0 5m 0 100n
.end
```